

High-Purity Germanium Detector Simulator with Radioactive Sources, Embedded in FPGA



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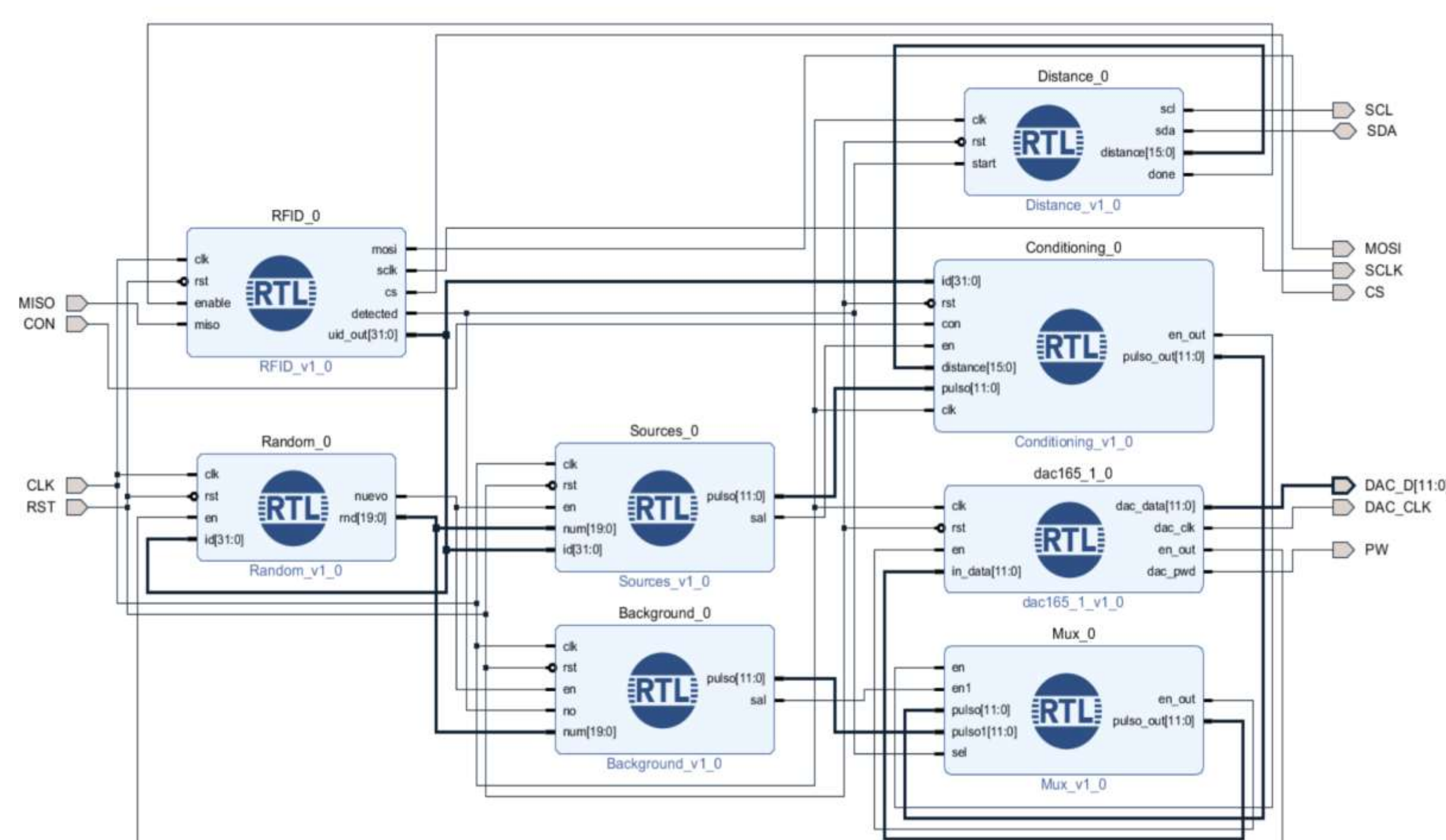
1. INSTRUCTIONS

The most reliable way to obtain accurate nuclear spectrometry data is through the use of **HPGe** detectors and radioactive sources, which provide excellent resolution and linear response [1-3]. However, their high cost, cryogenic requirements, and safety regulations often limit their use in educational and training laboratories [3]. As an alternative, **FPGA-based** simulators have been developed to reproduce nuclear pulse behavior in real time, offering safe and **cost-effective** tools for teaching and calibration [4,5]. Compared to Monte Carlo simulations, these systems provide interactive and experimental-like conditions [5].

In this work, we present an **FPGA-based** simulator integrated with **RFID** tokens and a **VL53LXX** sensor, capable of generating spectra conditioned by source type (**Cs-137**, **Co-60**, **Ba-133**, **Eu-152**, **Na-22**) and distance, and validated against experimental **HPGe** spectra [4,5]. This approach provides a flexible resource for nuclear instrumentation research and education.

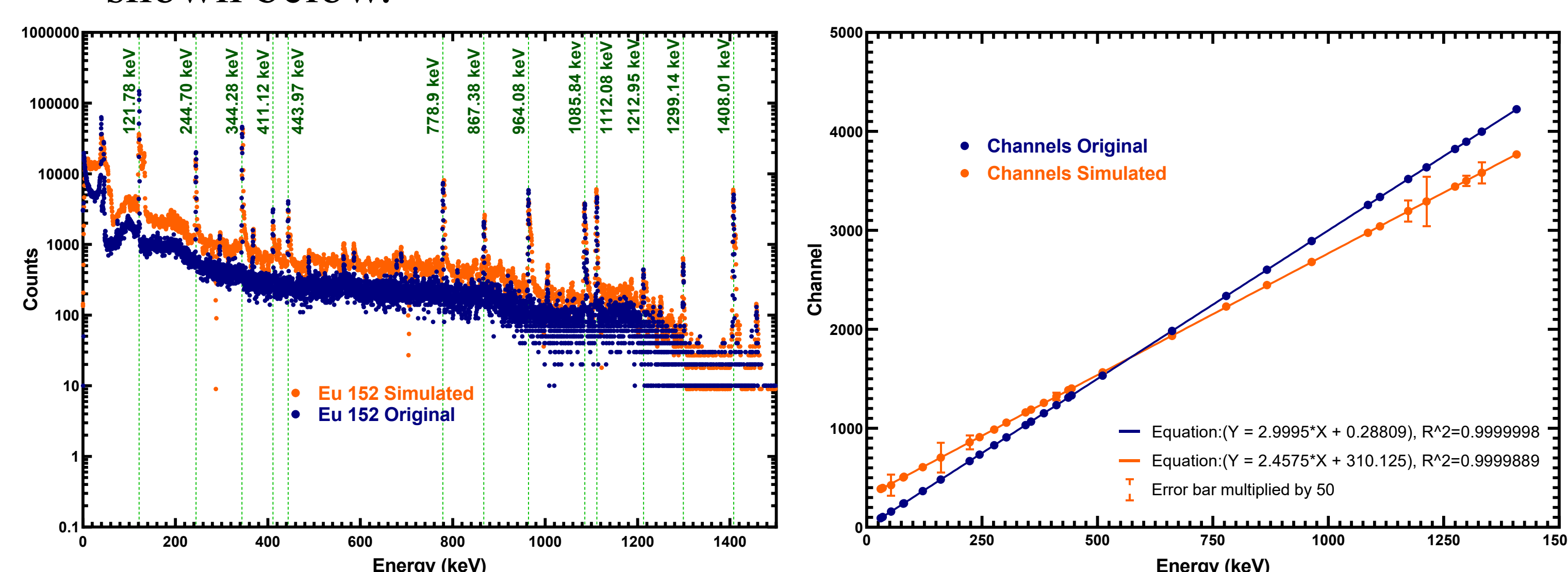
2. MATERIALS AND METHODS

For the detector simulation, an **RC522 RFID** reader module was implemented, with radioactive sources emulated through **RFID** tokens. Signal adjustment as a function of the source-detector distance was achieved using a **VL53LXX** distance sensor. Signal processing and conditioning were carried out on a **Zynq-7000 FPGA**, incorporating random number generation, source characteristics, and distance as key parameters. Furthermore, a **DAC902** digital-to-analog converter operating at **165 MSPS** with **12-bit** resolution was employed to synthesize trapezoidal pulses, following the configuration defined by the core architecture embedded in the **FPGA** platform.

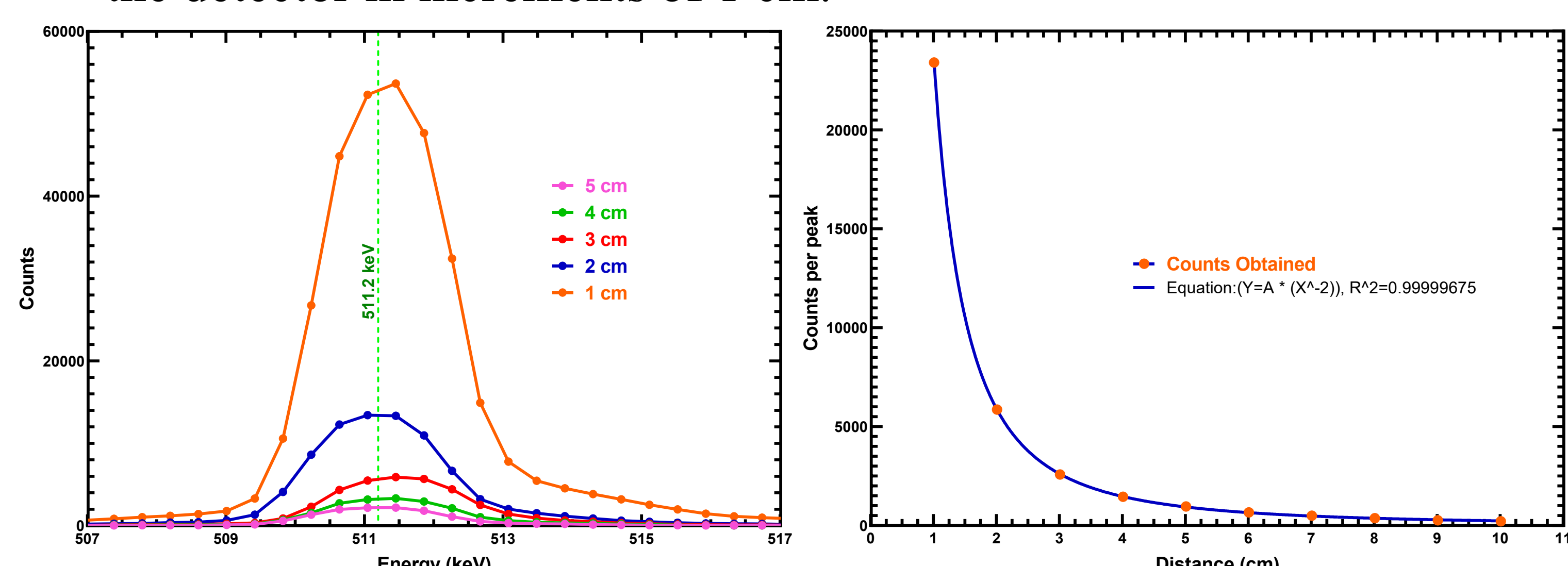


3. RESULTS

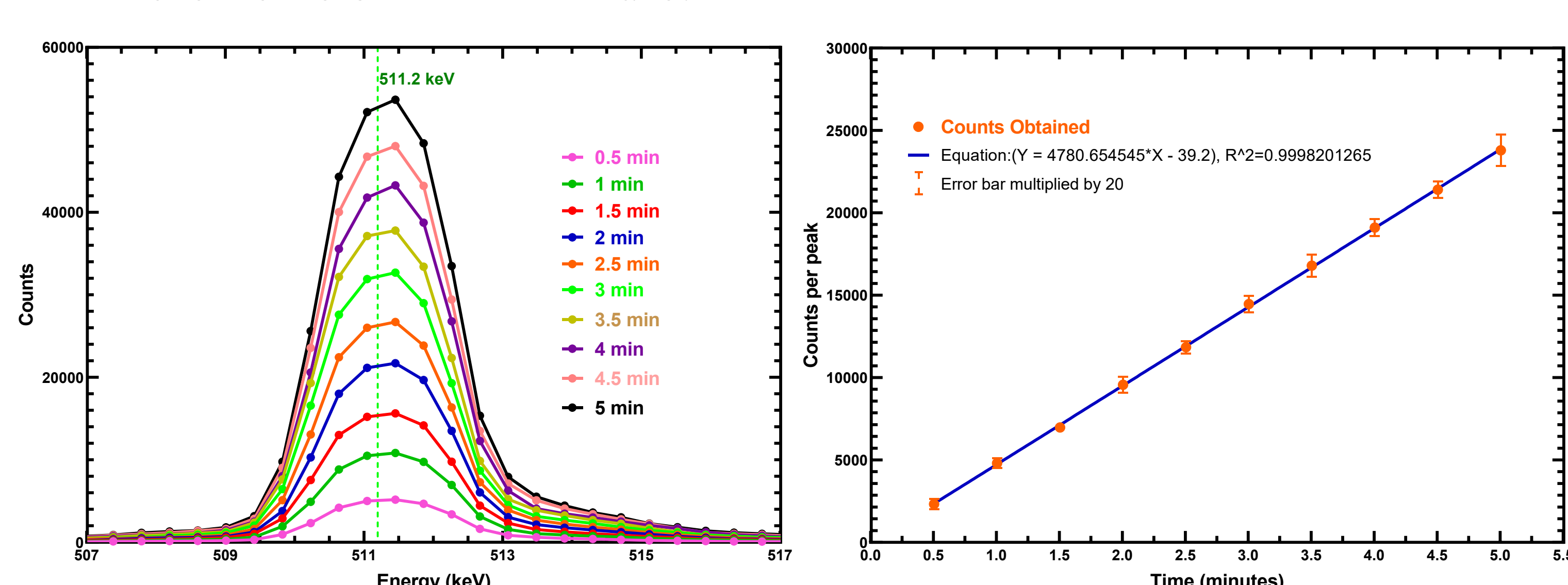
Spectra of **Cs-137**, **Co-60**, **Ba-133**, **Eu-152**, and **Na-22** were obtained. From these spectra, 31 characteristic **photopeaks** were identified, which were used to perform an energy-based calibration. The overlay of the spectra from the **Eu-152** source is shown below.



To verify the inverse square law of distance, the **Na-22** source was used, performing measurements from 1 cm to 10 cm away from the detector in increments of 1 cm.



Using the **Na-22** source, ten measurements were performed at ten different times, ranging from 0.5 minutes to 5 minutes in increments of half a minute.



4. CONCLUSIONS

The proposed **FPGA-based** system successfully emulates **HPGe** detector responses using **RFID**-tagged sources. The integration of **energy**, **time**, and **distance** calibrations ensures reliability for both nuclear instrumentation research and teaching. This approach provides a **cost-effective** and **flexible educational tool** for nuclear physics training.

5. REFERENCES

- [1] Rastgoo M, et al. Investigating the impact of source geometry and distance on HPGe detector efficiency using MCNPX simulations. 2024.
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- [3] Kaya S, et al. Effect of front, lateral and back dead layer thicknesses of a HPGe detector on full energy peak efficiency. 2022.
- [4] Zhang H, et al. Digital pulse processing for gamma-ray spectroscopy using FPGA. 2024.
- [5] Bohne W. et al. Teaching nuclear instrumentation with FPGA-based digital electronics. 2022.